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Top Skills

Spring Boot
Model Context Protocol (MCP)
Next.js

Languages

Ukrainian (Native or Bilingual)
English (Professional Working)

Oleksandr Zalizniak

CTO @ Signus.ai
United States

Summary

- Experienced Software Architect and Full Stack Developer with over 15 years of experience in designing, developing, and optimizing backend and full-stack solutions across various industries, including SaaS, e-commerce, finance, and telecommunications.
- Proven track record in leading complex projects, from conceptualization to deployment, with expertise in cloud infrastructure, microservices architecture, and DevOps practices. Responsible for leading infrastructure work and carrying out the technical specifications for various projects, ensuring alignment with business goals and technical requirements.
- Experienced in product engineering, with a strong passion for driving innovation and aligning technical solutions with product vision.
- Skilled in multiple programming languages and frameworks, including Java, Node.js, Spring and Nest.js, with a strong focus on scalability, performance, and user experience.
- Proven experience in both startup and corporate settings, effectively managing cross-functional teams to deliver high-quality solutions.
- Extensive experience with AWS, including infrastructure design, deployment automation, and cost optimization.
- Known for strong problem-solving skills, attention to detail, and a commitment to continuous improvement.

Experience

Signus.ai
Co-Founder & CTO
January 2025 - Present (1 year 5 months)
United States

As founding engineer and CTO of Signus.ai, I built the platform from zero — backend, AI service, frontend, and infrastructure. Signus.ai is an AI-

native contract intelligence platform powered by Sigi, our AI legal agent for generating, signing, analyzing, and tracking agreements at scale.

What I built:

- An LLM-powered agent runtime on Temporal — durable workflows for chat, redline, risk analysis, and contract Q&A, driving LLM calls with tool-use/ function-calling, retrieval, and structured output across TypeScript and Python workers.
- A model-agnostic LLM tier over AWS Bedrock + OpenAI behind a shared client — provider routing, prompt/version management, streaming, and tool-calling, so product code stays decoupled from any single model. Exposed internal capabilities (document search, contract data, signing actions) to the agent as MCP servers / tools, giving the LLM a typed interface to platform operations.
- A multi-app React frontend (Next.js monorepo, TypeScript) — main product app, marketing site, signing flow, admin dashboard, embedded HubSpot apps, and the real-time chat/redline UI for the AI agent.
- A polyglot backend — TypeScript (NestJS) for product logic, a Python (FastAPI) document-intelligence service (PDF processing, signature handling, text extraction, embeddings, field classification), and a Java / Spring Boot service for secure authentication.
- AWS infrastructure on ECS Fargate, provisioned via Terraform, with end-to-end observability (OpenTelemetry, CloudWatch, Sentry).

Stack: TypeScript, NestJS, React / Next.js, Java / Spring Boot, Python, FastAPI, LLM APIs (Anthropic Claude, OpenAI) via AWS Bedrock, MCP, Temporal, AWS (ECS, Lambda), Postgres, Elasticsearch, Terraform.

UserWay.org

3 years 10 months

Product Engineer - Confluence, Figma, Jira, Swagger, CI/CD
November 2023 - October 2024 (1 year)

Geneva, New York, United States

As the Product Engineer for Continuous Accessibility (<https://cicd.userway.org>) at UserWay, I led the design, development, and release of a cutting-edge tool that integrates accessibility checks into the CI/CD pipeline, allowing developers to detect and address accessibility issues early in the development process. I was responsible for driving the product from concept to launch, collaborating closely with cross-functional teams to ensure seamless integration with existing UserWay services.

Key Achievements:

- Developed and implemented CI/CD flows, merging scanning and E2E tests into a unified product.
- Designed and structured large-scale Nest.js application
- Collaborated with the design team to create wireframes and implement design standards, ensuring the product met the highest quality standards.
- Worked extensively on public documentation, API packages, and CLI kits, providing comprehensive tools for developers.
- Conducted in-depth competitor analysis and adjusted the product strategy to maintain a competitive edge.
- Managed a diverse team of developers, frontend engineers, and QA automation specialists, guiding them through the product lifecycle.

This role has solidified my expertise in product management, architecture design, and cross-functional team leadership, enabling me to deliver a product that significantly enhances web accessibility for large organizations.

Lead Backend Architect - AWS, ECS, Lambda, Kafka, Postgresql, Redis, Bitbucket, Java, Node.js, React
January 2021 - October 2023 (2 years 10 months)
Geneva, New York, United States

As the Lead Backend Architect at UserWay, I was responsible for designing and implementing cloud infrastructure on AWS and developing backend architecture from scratch. My role extended into product development, where I gained valuable insights into SaaS operations.

Key Responsibilities:

- Cloud Infrastructure: Architected and managed AWS infrastructure, including ECS, Batch, Redis, RDS, and AWS Lambda, handling high traffic loads with nearly 10,000 requests per second at peak times.
- Microservices Architecture: Developed and maintained over 15 active microservices, with a focus on scalability, high availability, and efficient communication.
- Backend Development: Led backend development using Node.js and Java, with MySQL and MongoDB for data management, ensuring robust and scalable microservices.
- Frontend Integration: Collaborated with frontend teams (React/Angular), deploying applications as S3 web apps with CDN integration.

- Performance Monitoring: Implemented metrics for latency, autoscaling, and service availability, leveraging AWS tools for real-time optimization.
- Security & Compliance: Enhanced security measures, including managing a significant early incident and ongoing reviews to meet compliance standards like GDPR.
- Product Development: Contributed to strategic decisions, including the development of the Accessibility Scanner

This role allowed me to apply my expertise in cloud architecture and microservices while broadening my experience in SaaS product development, ultimately supporting UserWay's mission to enhance web accessibility.

Crazy Labs

Backend Developer - AWS, PostgreSQL, Redis, ECS, Jenkins, Node.js, Java

October 2019 - January 2021 (1 year 4 months)

Kyiv

At Crazy Labs, I initially joined as a Java developer and then transitioned to Node.js, taking on the role of Lead Backend Developer for the <https://developers.crazylabs.com> product. In this role, I was responsible for setting up the backend architecture, leading major refactoring efforts, and significantly improving test coverage to ensure the stability and scalability of the platform.

Key Achievements:

- Backend Architecture: Designed and implemented a cloud-based backend architecture on AWS, utilizing PostgreSQL, Redis, and ECS to ensure a robust and scalable foundation.
- Technology Transition: Transitioned from Java to Node.js, effectively leading the backend development for the project.
- Refactoring: Conducted extensive refactoring to optimize codebase performance and maintainability.
- Test Coverage: Enhanced test coverage across the backend, leading to improved code quality and reduced production issues.

This role allowed me to refine my skills in backend development, architecture design, and cloud-based infrastructure, contributing to the reliability of the Crazy Labs developer portal.

Platin.io

Full Stack Developer - AWS, DynamoDb, Redis, RDS, S3, CloudFront
May 2017 - October 2019 (2 years 6 months)

Kyiv

At Platin.io, a blockchain-related startup, I worked as a Full Stack Engineer and DevOps, where I was responsible for leading the infrastructure work and executing the technical specifications for the Platin project. My role encompassed the architecture, design, implementation, and deployment of all back-end components, along with setting up and configuring a new AWS account, including build automation and autoscaling.

Key Achievements:

- Infrastructure Leadership: Led the creation and management of the AWS cloud infrastructure, utilizing services such as VPC, EC2, ECS, ELB, DynamoDb, Redis, RDS, S3, and CloudFront, ensuring scalability and performance.
- Full Stack Development: Designed and implemented backend components using Spring Boot, JPA, Java 8, and Postgres, while also working on the frontend with Angular and Docker.
- API Development: Developed and documented APIs using Swagger, contributing to a robust and efficient distributed system.
- Cloud Automation: Set up build automation and autoscaling, enhancing the deployment process and system reliability.

Despite the eventual closure of Platin.io, this role allowed me to gain deep experience in full stack development, cloud infrastructure, and blockchain technology, providing valuable insights into the challenges and opportunities of working in a startup environment.

XPlace.com

Full Stack Engineer - AWS, Java, Spring
February 2015 - October 2019 (4 years 9 months)

Israel

At XPlace.com, I was responsible for the infrastructure, DevOps, and development of a leading freelance marketplace, while also contributing to the development of UserWay.org. My role involved creating, configuring, and administering AWS accounts for both projects, where I played a pivotal role in architecture, design, and implementation.

Key Achievements:

- Infrastructure & DevOps: Led the creation and management of AWS infrastructure, including services such as VPC, EC2, ECS, ELB, Redis, RDS, S3, and CloudFront, ensuring reliable and scalable deployments.
- Project Development: Developed key components for the XPlace marketplace using Java, Spring, Hibernate, and AngularJS, while also contributing to Node.js development for UserWay.org.
- Cloud Migration: Spearheaded the migration from dedicated hardware (SoftLayer) to AWS, transitioning from MVC to SPA and web services architecture, which significantly improved user experience and system efficiency.
- Performance Optimization: Optimized performance, configured auto-scaling, and implemented AWS cost optimization strategies for UserWay.org, resulting in better resource management and reduced operational costs.
- Deployment Automation: Configured and maintained deployment automation processes, streamlining the release pipeline and ensuring smooth updates.

This experience provided me with deep insights into full stack development, cloud infrastructure management, and DevOps, all of which contributed to the success and scalability of these two significant projects.

Nabus Development

Full Stack Engineer - Java, Spring, Tomcat, JSP, Angular.JS

February 2015 - July 2015 (6 months)

Kyiv

At Nabus Development, I was responsible for maintaining and enhancing an existing MVC application. My key contribution was transitioning the application to a Single Page Application (SPA), which significantly improved the user experience and performance. I also worked on optimizing the backend processes to support the new frontend architecture, ensuring seamless interaction between the client and server.

Key Responsibilities:

- MVC Maintenance: Managed and maintained the existing MVC-based web application, ensuring its stability and performance.
- SPA Transition: Led the transition from an MVC structure to a Single Page Application (SPA), enhancing user experience and responsiveness.

- Backend Optimization: Optimized backend processes to align with the SPA architecture, improving data handling and communication between client and server.
- User Experience Improvement: Focused on enhancing the user interface and overall experience by implementing modern frontend technologies.

This role allowed me to apply my expertise in both frontend and backend development, contributing to a smoother, more efficient user experience.

EPAM Systems

Software Engineer - eCommerce, Hybris, Java

September 2014 - January 2015 (5 months)

Kyiv

As a key Hybris developer on the Travis Perkins project at EPAM Systems, I was responsible for developing a mobile version of a B2B application, with a focus on CMS and UI customization. My role involved collaborating closely with the team to deliver high-quality solutions that met the client's requirements.

Key Responsibilities:

- Estimations & Planning: Involved in the estimation and planning process, contributing to accurate project timelines and resource allocation.
- Implementation Review: Conducted reviews of implementation to ensure adherence to best practices and technical standards.
- Technical Discussions: Engaged in discussions on technical approaches and solutions, ensuring that the most effective strategies were employed.
- Business Analysis Communication: Maintained regular communication with Business Analysts to discuss status updates, changes, re-estimations, and acceptance criteria.
- Cross-Team Collaboration: Facilitated communication with other teams and technical leads to share information and align on project goals.

Environment:

This role allowed me to deepen my expertise in Hybris development while contributing to a high-impact project that enhanced the mobile capabilities of a leading B2B application.

Trinetix LLC

Fullstack Java developer - Java, Spring

May 2014 - August 2014 (4 months)

Kyiv / remote job

As a Java/J2EE Developer at Trinetix, I was responsible for the development of a web portal for a Ukrainian media company. My role involved the design and creation of the database schema, as well as the application design and development based on the Spring framework.

Key Responsibilities:

- DB Schema Design: Designed and created the database schema to ensure efficient data storage and retrieval.
- Application Development: Developed the web application using the Spring framework (Spring MVC), ensuring a robust and scalable solution.
- Technology Stack: Worked with a range of technologies, including Hibernate for ORM, Eclipse for development, Java 8 for backend programming, and jQuery for frontend interactions.

This project allowed me to apply my expertise in Java and J2EE development to create a high-performing web portal, contributing to the media company's digital presence.

GlobalLogic

Java Developer - Java, Desktop app

April 2012 - May 2014 (2 years 2 months)

Kyiv, Ukraine

At GlobalLogic, I worked on the Avid Interplay Capture project, focusing on developing a client-server application that facilitates the capturing and processing of media. My role involved both feature implementation and addressing technical challenges to meet the project's requirements.

Key Responsibilities:

- Feature Implementation: Developed and implemented new features for the application, enhancing its capabilities for media capture and processing.
- Planning & Estimation: Participated in project planning and estimation to ensure accurate timelines and resource allocation.
- Bug Fixing: Identified and resolved bugs, improving the application's stability and performance.
- Reverse Engineering: Conducted reverse engineering to understand existing codebases and integrate new functionalities effectively.

This experience enabled me to refine my skills in Java development and contribute to a complex media processing application, playing a key role in its ongoing improvement and feature expansion.

Lizard Soft

Java, J2EE Developer - Java, EJB, JPA, WebSphere

August 2012 - December 2012 (5 months)

Kyiv, Ukraine

At Lizard Soft, I contributed to the development of a banking system built on the J2EE technology stack. My role involved designing and implementing key components that ensured the system's integration and operational efficiency.

Key Responsibilities:

- DB Schema Design: Created and optimized the database schema to support the banking system's data requirements.
- Integration Layer Implementation: Developed an integration layer to connect the legacy web application with an external system, facilitating seamless data exchange.
- Workflow Design: Designed workflows based on JMS (Java Message Service), ensuring reliable messaging and process automation within the system.

This project allowed me to apply my expertise in J2EE development, contributing to a critical banking system with a focus on integration, workflow automation, and database management.

HP Enterprise Services

ATG developer - ATG, JBoss, Oracle

September 2011 - December 2011 (4 months)

Cairo, Egypt

At HP Enterprise Services, I contributed to the development of Vodafone's eShop (vodafone.nl), an e-commerce web application built on the ATG platform. My role involved designing and implementing custom functionality to enhance the application's inventory management and stock information features.

Key Responsibilities:

- Design Documentation: Developed detailed design documents to guide the implementation of new features and ensure alignment with project goals.
- Custom Functionality Implementation: Implemented custom stock information-related functionality, leveraging the ATG platform's inventory capabilities.
- Automated Testing: Created automated test cases to ensure the reliability and accuracy of the new features.

This role allowed me to refine my skills in ATG development and contribute to a high-profile e-commerce application, delivering robust and scalable solutions for Vodafone Netherlands.

Lizard Soft

Java, J2EE Developer - Spring, JPA, JMS, WebSphere, Oracle
April 2011 - August 2011 (5 months)

Kyiv, Ukraine

At Lizard Soft, I worked remotely on the AG project, a backend system dedicated to the accounting and automation of operations involving precious metals. This complex system provided an administrative GUI and a JMS interface for external applications, with over 100 operations implemented.

Key Responsibilities:

- Database Structure Design: Created the database structure using Enterprise Architect to support the system's extensive operations.
- XML Messaging Design: Designed the structure of XML messages for each operation, defining the API for communication with external systems.
- Operation Implementation: Developed and implemented operations by creating message beans for JMS, mapping to XML, and writing message processors. Conducted thorough integration tests to ensure system reliability.
- Client Collaboration: Regularly clarified specifications and requirements with the customer (Web Format LLC) to ensure the system met business needs.

This role allowed me to apply my expertise in Java and J2EE development to a highly specialized system, ensuring its robustness and functionality in automating complex operations.

EPAM Systems

ATG Developer - Java, ATG, Weblogic
March 2007 - March 2011 (4 years 1 month)

Kyiv, Ukraine

At EPAM Systems, I worked on several high-impact projects, including O2 Web Self Care, Screwfix.com, Trade Counter, and NexDayDIY.com, where I played a crucial role in architectural design, implementation, and customer communication. My responsibilities spanned across multiple aspects of project development, from detailed design documentation to mentoring new team members.

Key Responsibilities:

- O2 Web Self Care (moje.o2.cz): Created detailed use case architectural documentation, clarified requirements with the customer, and aligned on technical implementations. Assisted with environment setup and ensured design documents were used effectively by the development team.
- Screwfix.com: Provided support for the legacy application, implemented minor functionality and appearance changes, and ensured protection against XSS vulnerabilities. Led the refactoring and restructuring of the existing codebase, created new module hierarchies, and implemented frontend and third-party integrations. Managed the acceptance environment and optimized the build and deployment processes.
- Trade Counter: Focused on frontend development, integration with web services, and improving the efficiency of ATG applications. Refactored Fitness test codebase and provided architectural solutions.
- NexDayDIY.com: Set up the deployment environment, created application module structures, implemented custom business and presentation logic, and ensured SEO optimization for site navigation.

This role allowed me to deepen my expertise in Java, J2EE, and ATG development while contributing to complex e-commerce and enterprise applications, ensuring their scalability, performance, and alignment with customer needs.

Ukrtelecom

.Net software developer - IIS, MS Visual Studio, .Net
October 2005 - March 2007 (1 year 6 months)

Kherson, Ukraine

At Ukrtelekom, I was responsible for developing and maintaining several critical projects that facilitated the interaction between the company's Billing and Technical Accounting systems. My work focused on designing and

implementing web services, developing desktop and web applications, and ensuring seamless synchronization between heterogeneous systems.

Key Responsibilities:

- ASKR Web Service: Developed a .NET web service for IIS based on WSDL specifications, facilitating data representation from the Billing system. Designed Oracle database structures, created stored procedures, and ensured efficient interaction between the web service and the database using ADO.NET.
- Interops: Implemented a .NET CLI program to synchronize the Billing and Technical Accounting systems using the ASKR and ASTLU web services. Developed business logic, implemented web service clients, and conducted system tests to ensure data consistency.
- Verificator: Developed a "Win Form" application to visualize and control the synchronization process between ASKR and ASTLU. Designed the GUI, implemented multi-threading for efficient processing, and tested the application for reliability.
- ORDERS.NET: Created a web application for processing customer orders at telephone stations. Designed the database structure, developed and tested the web service, and contributed to the design of the web client part using ASP.NET, HTML/DHTML, JavaScript, and CSS.

This role allowed me to apply my expertise in .NET development, database design, and system integration, contributing to the efficiency and reliability of Ukrtelekom's operational systems.

Education

Kherson State University

Bachelor, Mathematics, Physics and Informatics · (1997 - 2002)

Kherson National Technical University

Bachelor, Computer Sciences, Application Software
Development · (2004 - 2005)